

Speed for Maths

Increase your calculation speed by up to 30% through daily practice for 20 minutes ($\approx 0.3h$).

1. Squares: 1 to 50
2. $\frac{___}{***x***}$: 1 to 50
3. Cubes: 1 to 30
4. % Value: 50+
5. Table: 1-30
6. Powers of 2: 2^2 2^{16}
7. Multiplication values
8. Powers of 3 & 5 (up to 3^{10} and 5^{10})
9. **Factorials**: $1!$ to $10!$
10. Prime numbers: Up to 100

Daily Quiz

- <https://github.com/mohitdhami/Speed-for-Maths>

(1) Squares: 1 to 50

Squares 1 to 50 Chart

$1^2 = 1$	$13^2 = 169$	$26^2 = 676$	$39^2 = 1521$
$2^2 = 4$	$14^2 = 196$	$27^2 = 729$	$40^2 = 1600$
$3^2 = 9$	$15^2 = 225$	$28^2 = 784$	$41^2 = 1681$
$4^2 = 16$	$16^2 = 256$	$29^2 = 841$	$42^2 = 1764$
$5^2 = 25$	$17^2 = 289$	$30^2 = 900$	$43^2 = 1849$
$6^2 = 36$	$18^2 = 324$	$31^2 = 961$	$44^2 = 1936$
$7^2 = 49$	$19^2 = 361$	$32^2 = 1024$	$45^2 = 2025$
$8^2 = 64$	$20^2 = 400$	$33^2 = 1089$	$46^2 = 2116$
$9^2 = 81$	$21^2 = 441$	$34^2 = 1156$	$47^2 = 2209$
$10^2 = 100$	$22^2 = 484$	$35^2 = 1225$	$48^2 = 2304$
$11^2 = 121$	$23^2 = 529$	$36^2 = 1296$	$49^2 = 2401$
$12^2 = 144$	$24^2 = 576$	$37^2 = 1369$	$50^2 = 2500$
	$25^2 = 625$	$38^2 = 1444$	

(2) ___/***x***: 1 to 50

#1

Square Roots (1 to 25)

$$\sqrt{1} = 1$$

$$\sqrt{2} \sim 1.41$$

$$\sqrt{3} \sim 1.73$$

$$\sqrt{4} = 2$$

$$\sqrt{5} \sim 2.24$$

$$\sqrt{6} \sim 2.45$$

$$\sqrt{7} \sim 2.65$$

$$\sqrt{8} \sim 2.83$$

$$\sqrt{9} = 3$$

$$\sqrt{10} \sim 3.16$$

$$\sqrt{11} \sim 3.32$$

$$\sqrt{12} \sim 3.46$$

$$\sqrt{13} \sim 3.61$$

$$\sqrt{14} \sim 3.74$$

$$\sqrt{15} \sim 3.87$$

$$\sqrt{16} = 4$$

$$\sqrt{17} \sim 4.12$$

$$\sqrt{18} \sim 4.24$$

$$\sqrt{19} \sim 4.36$$

$$\sqrt{20} \sim 4.47$$

$$\sqrt{21} \sim 4.58$$

$$\sqrt{22} \sim 4.69$$

$$\sqrt{23} \sim 4.80$$

$$\sqrt{24} \sim 4.90$$

$$\sqrt{25} = 5$$

#2

Square Roots (26 to 50)

$$\sqrt{26} \sim 5.10$$

$$\sqrt{27} \sim 5.20$$

$$\sqrt{28} \sim 5.29$$

$$\sqrt{29} \sim 5.39$$

$$\sqrt{30} \sim 5.48$$

$$\sqrt{31} \sim 5.57$$

$$\sqrt{32} \sim 5.66$$

$$\sqrt{33} \sim 5.74$$

$$\sqrt{34} \sim 5.83$$

$$\sqrt{35} \sim 5.92$$

$$\sqrt{36} = 6$$

$$\sqrt{37} \sim 6.08$$

$$\sqrt{38} \sim 6.16$$

$$\sqrt{39} \sim 6.24$$

$$\sqrt{40} \sim 6.32$$

$$\sqrt{41} \sim 6.40$$

$$\sqrt{42} \sim 6.48$$

$$\sqrt{43} \sim 6.56$$

$$\sqrt{44} \sim 6.63$$

$$\sqrt{45} \sim 6.71$$

$$\sqrt{46} \sim 6.78$$

$$\sqrt{47} \sim 6.86$$

$$\sqrt{48} \sim 6.93$$

$$\sqrt{49} = 7$$

$$\sqrt{50} \sim 7.07$$

(3) Cubes: 1 to 30

$$1^3 = 1$$

$$2^3 = 8$$

$$3^3 = 27$$

$$4^3 = 64$$

$$5^3 = 125$$

$$6^3 = 216$$

$$7^3 = 343$$

$$8^3 = 512$$

$$9^3 = 729$$

$$10^3 = 1000$$

$$11^3 = 1331$$

$$12^3 = 1728$$

$$13^3 = 2197$$

$$14^3 = 2744$$

$$15^3 = 3375$$

$$16^3 = 4096$$

$$17^3 = 4913$$

$$18^3 = 5832$$

$$19^3 = 6859$$

$$20^3 = 8000$$

$$21^3 = 9261$$

$$22^3 = 10648$$

$$23^3 = 12167$$

$$24^3 = 13824$$

$$25^3 = 15625$$

$$26^3 = 17576$$

$$27^3 = 19683$$

$$28^3 = 21952$$

$$29^3 = 24389$$

$$30^3 = 27000$$

(4) % Value: 50+

#1

Important fractional values.

$$\cdot 27.27\% = 27\frac{3}{11}\% = 3(9\frac{1}{11}) = 3(9.09\%) = 3 \times \frac{1}{11} = \frac{3}{11}$$

$$\cdot 45.45\% = 45\frac{5}{11}\% = 5(9\frac{1}{11}) = 5(9.09\%) = 5 \times \frac{1}{11} = \frac{5}{11}$$

$$\cdot 63.63\% = 63\frac{7}{11}\% = 7(9\frac{1}{11}) = 7(9.09\%) = 7 \times \frac{1}{11} = \frac{7}{11}$$

$$\uparrow [9.09\% = \frac{1}{11}]$$

#2

$\cdot \frac{1}{2} = 50\%$	$\cdot \frac{1}{7} = 14\frac{2}{7}\% = 14.28\%$
$\cdot \frac{1}{4} = 25\%$	$\cdot \frac{1}{14} = 7\frac{1}{7}\% = 7.14\%$
$\cdot \frac{1}{8} = 12.5\%$	$\cdot \frac{2}{7} = 28\frac{4}{7}\% = 28.56\%$
$\cdot \frac{1}{16} = 6.25\%$	$\cdot \frac{3}{7} = 42\frac{6}{7}\%$
$\cdot \frac{1}{32} = 3.125\%$	$\cdot \frac{4}{7} = 56\frac{8}{7}\% = 57\frac{1}{7}\%$
	$\cdot \frac{5}{7} = 70\frac{10}{7}\% = 71\frac{3}{7}\%$

#3

Fraction to Percentage ($9 \neq 11$)

$$\cdot \frac{1}{9} = 11 \frac{1}{9} \% = 11.11\%$$

$$\cdot \frac{1}{11} = 9 \frac{1}{11} \% = 9.09\%$$

$$\cdot \frac{2}{9} = 22 \frac{2}{9} \% = 22.22\%$$

$$\cdot \frac{3}{11} = 27 \frac{3}{11} \% = 27.27\%$$

$$\frac{4}{9} = 44 \frac{4}{9} \% = 44.44\%$$

$$\cdot \frac{5}{11} = 45 \frac{5}{11} \% = 45.45\%$$

$$\frac{7}{9} = 77 \frac{7}{9} \% = 77.77\%$$

$$\cdot \frac{2}{11} = 18 \frac{2}{11} \% = 18.18\%$$

Fraction to Percentage (Multiple of 8)

$$\frac{1}{8} = 12.5\%$$

$$\cdot \frac{3}{8} = 37.5\% = 37 \frac{1}{2}\%$$

$$\frac{1}{16} = 6.25\%$$

$$\cdot \frac{5}{8} = 62.5\% = 62 \frac{1}{2}\%$$

= More Important Fractional values

$$81.81\% = \frac{9}{11}$$

$$\cdot 54.54\% = \frac{6}{11}$$

$$33.33\% = 33 \frac{1}{3} \% = \frac{3}{9} = \frac{1}{3}$$

$$44.44\% = 44 \frac{4}{9} \% = \frac{4}{9}$$

$$55.55\% = 55 \frac{5}{9} \% = \frac{5}{9}$$

$$77.77\% = 77 \frac{7}{9} \% = \frac{7}{9}$$

$$66.66\% = 66 \frac{2}{3} \% = \frac{2}{3}$$

$$\frac{1}{7} = 14 \frac{2}{7} \% = 14.28\%$$

#4

$$\bullet \quad 28 \frac{4}{7} \% = 28,56 \% = \frac{2}{7}$$

$$\bullet \quad 42 \frac{6}{7} \% = 42,84 \% = \frac{3}{7}$$

$$\bullet \quad 57 \frac{1}{7} \% = 57,12 \% = \frac{4}{7}$$

$$\bullet \quad 71 \frac{3}{7} \% = 71,40 \% = 70 \frac{10 \frac{1}{3}}{7} = 71 \frac{3}{7} \% \\ \begin{array}{c} \swarrow \quad \searrow \\ 14 \times 5 \quad 2 \times 5 \end{array} \\ = \frac{5}{7}$$

#,

$$\bullet \quad 12 \frac{1}{2} \% = 12,5 \% = \frac{1}{8}$$

$$\bullet \quad 25 \% = \frac{1}{4}$$

$$\bullet \quad 37 \frac{1}{2} \% = 37,5 \% = 50 \% - 12,5 \% = \frac{1}{2} - \frac{1}{8} = \frac{4-1}{8} = \frac{3}{8}$$

$$\bullet \quad 62 \frac{1}{2} \% = 62,5 \% = 50 \% + 12,5 \% = \frac{1}{2} + \frac{1}{8} = \frac{4+1}{8} = \frac{5}{8}$$

$$\bullet \quad 87 \frac{1}{2} \% = 87,5 \% = 100 \% - 12,5 \% = 1 - \frac{1}{8} = \frac{8-1}{8} = \frac{7}{8}$$

$$\hookrightarrow \text{OR, } 50 \% + 37,5 \% = \frac{1}{2} + (12,5 \% \times 3) = \frac{1}{2} + \frac{3}{8} = \frac{4+3}{8} = \frac{7}{8}$$

$$\bullet \quad 212 \frac{1}{2} \% = 212,5 \% = 200 \% + 12,5 \% = 2 + \frac{1}{8} = \frac{17}{8}$$

$$\bullet \quad 116 \frac{2}{3} \% = 100 \% + (8,33 \% \times 2) = 1 + \frac{2}{3} = \frac{5}{3}$$

$$\bullet \quad 128 \frac{4}{7} \% = 128,56 \% = 100 \% + (14,28 \% \times 2) = 1 + \frac{2}{7} = \frac{9}{7}$$

$$\bullet \quad 208 \frac{1}{3} \% = 208,33 \% = 200 \% + 8,33 \% = 2 + \frac{1}{12} = \frac{25}{12}$$

#5

Fraction to Percentage $\Rightarrow$

$$\cdot \frac{100}{100} \times 100 = 100\%$$

$$\# \left[x\% = \frac{x}{100} \right]$$

$$\cdot \frac{50}{100} \times 100 = 50\%$$

$$\cdot \frac{25}{100} \times 100 = 25\%$$

$$\cdot \frac{1}{7} \times 100 = 14\frac{2}{7}\% = 14.28\%$$

$$\cdot \frac{1}{6} \times 100 = 16\frac{4}{3}\% = 16.66\%$$

$$\cdot 20\% = \frac{20}{100} = \frac{1}{5}$$

$$\cdot 30\% = \frac{30}{100} = \frac{3}{10}$$

$$\cdot 25\% = \frac{25}{100} = \frac{1}{4}$$

$$\cdot 66\frac{2}{3}\% = \frac{200}{3} \times \frac{1}{100} = \frac{2}{3}$$

$$16.67\% = 16\frac{2}{3}\% = \frac{50}{3} \times \frac{1}{100} = \frac{1}{6}$$

$$12.5\% = 12\frac{1}{2}\% = \frac{25}{2} \times \frac{1}{100} = \frac{1}{8}$$

$$\frac{1}{2} = 50\%$$

$$\cdot \frac{1}{6} = 16\frac{2}{3}\% = 16.66\%$$

$$\cdot \frac{1}{13} = 7\frac{9}{13}\% = 7.69\%$$

$$\frac{1}{2}$$

$$\cdot \frac{1}{7} = 14\frac{2}{7}\% = 14.28\%$$

$$\cdot \frac{1}{14} = 7\frac{2}{7}\% = 7.14\%$$

$$\frac{1}{3} = 33.3\% = 33\frac{1}{3}$$

$$\cdot \frac{1}{8} = 12\frac{1}{2}\% = 12.5\%$$

$$\cdot \frac{1}{15} = 6\frac{2}{3}\% = 6.66\%$$

$$\frac{1}{3}$$

$$\cdot \frac{1}{9} = 11\frac{1}{9}\% = 11.11\%$$

$$\cdot \frac{1}{16} = 6\frac{1}{4}\% = 6.25\%$$

$$\frac{1}{4} = 25\%$$

$$\cdot \frac{1}{10} = 10\%$$

$$\cdot \frac{1}{17} = 5\frac{15}{17}\% = 5.88\%$$

$$\frac{1}{4}$$

$$\cdot \frac{1}{11} = 9\frac{1}{11}\% = 9.09\%$$

$$\cdot \frac{1}{18} = 5.55\%$$

$$\frac{1}{5} = 20\%$$

$$\cdot \frac{1}{12} = 8\frac{1}{3}\% = 8.33\%$$

$$\cdot \frac{1}{19} = 5\frac{5}{19}\% = 5.26\%$$

#6

$\frac{1}{1} = 100\%$	$\frac{1}{11} = 9.09\%$	$\frac{4}{6} = 66.67\%$	$\frac{5}{11} = 45.45\%$
$\frac{1}{2} = 50\%$	$\frac{1}{12} = 8.33\%$	$\frac{5}{7} = 71.43\%$	$\frac{7}{11} = 63.64\%$
$\frac{1}{3} = 33.33\%$	$\frac{2}{3} = 66.67\%$	$\frac{6}{7} = 85.71\%$	$\frac{10}{11} = 90.91\%$
$\frac{1}{4} = 25\%$	$\frac{3}{4} = 75\%$	$\frac{2}{8} = 25\%$	$\frac{5}{12} = 41.67\%$
$\frac{1}{5} = 20\%$	$\frac{2}{5} = 40\%$	$\frac{3}{8} = 37.5\%$	$\frac{7}{12} = 58.33\%$
$\frac{1}{6} = 16.67\%$	$\frac{3}{5} = 60\%$	$\frac{5}{8} = 62.5\%$	$\frac{11}{12} = 91.67\%$
$\frac{1}{7} = 14.29\%$	$\frac{4}{5} = 80\%$	$\frac{7}{8} = 87.5\%$	$\frac{1}{13} = 7.69\%$
$\frac{1}{8} = 12.5\%$	$\frac{5}{6} = 83.33\%$	$\frac{2}{9} = 22.22\%$	$\frac{1}{14} = 7.14\%$
$\frac{1}{9} = 11.11\%$	$\frac{3}{6} = 50\%$	$\frac{7}{9} = 77.77\%$	$\frac{1}{16} = 6.25\%$
$\frac{1}{10} = 10\%$			$\frac{1}{24} = 4.17\%$
			$\frac{1}{25} = 4\%$
			$\frac{1}{30} = 3.33\%$

(5) Table: 12-30

#1

12

$12 \times 1 = 12$
 $12 \times 2 = 24$
 $12 \times 3 = 36$
 $12 \times 4 = 48$
 $12 \times 5 = 60$
 $12 \times 6 = 72$
 $12 \times 7 = 84$
 $12 \times 8 = 96$
 $12 \times 9 = 108$
 $12 \times 10 = 120$

13

$13 \times 1 = 13$
 $13 \times 2 = 26$
 $13 \times 3 = 39$
 $13 \times 4 = 52$
 $13 \times 5 = 65$
 $13 \times 6 = 78$
 $13 \times 7 = 91$
 $13 \times 8 = 104$
 $13 \times 9 = 117$
 $13 \times 10 = 130$

14

$14 \times 1 = 14$
 $14 \times 2 = 28$
 $14 \times 3 = 42$
 $14 \times 4 = 56$
 $14 \times 5 = 70$
 $14 \times 6 = 84$
 $14 \times 7 = 98$
 $14 \times 8 = 112$
 $14 \times 9 = 126$
 $14 \times 10 = 140$

15

$15 \times 1 = 15$
 $15 \times 2 = 30$
 $15 \times 3 = 45$
 $15 \times 4 = 60$
 $15 \times 5 = 75$
 $15 \times 6 = 90$
 $15 \times 7 = 105$
 $15 \times 8 = 120$
 $15 \times 9 = 135$
 $15 \times 10 = 150$

16

$16 \times 1 = 16$
 $16 \times 2 = 32$
 $16 \times 3 = 48$
 $16 \times 4 = 64$
 $16 \times 5 = 80$
 $16 \times 6 = 96$
 $16 \times 7 = 112$
 $16 \times 8 = 128$
 $16 \times 9 = 144$
 $16 \times 10 = 160$

17

$17 \times 1 = 17$
 $17 \times 2 = 34$
 $17 \times 3 = 51$
 $17 \times 4 = 68$
 $17 \times 5 = 85$
 $17 \times 6 = 102$
 $17 \times 7 = 119$
 $17 \times 8 = 136$
 $17 \times 9 = 153$
 $17 \times 10 = 170$

18

$18 \times 1 = 18$
 $18 \times 2 = 36$
 $18 \times 3 = 54$
 $18 \times 4 = 72$
 $18 \times 5 = 90$
 $18 \times 6 = 108$
 $18 \times 7 = 126$
 $18 \times 8 = 144$
 $18 \times 9 = 162$
 $18 \times 10 = 180$

19

$19 \times 1 = 19$
 $19 \times 2 = 38$
 $19 \times 3 = 57$
 $19 \times 4 = 76$
 $19 \times 5 = 95$
 $19 \times 6 = 114$
 $19 \times 7 = 133$
 $19 \times 8 = 152$
 $19 \times 9 = 171$
 $19 \times 10 = 190$

20

$20 \times 1 = 20$
 $20 \times 2 = 40$
 $20 \times 3 = 60$
 $20 \times 4 = 80$
 $20 \times 5 = 100$
 $20 \times 6 = 120$
 $20 \times 7 = 140$
 $20 \times 8 = 160$
 $20 \times 9 = 180$
 $20 \times 10 = 200$

#2

21

$21 \times 1 = 21$
 $21 \times 2 = 42$
 $21 \times 3 = 63$
 $21 \times 4 = 84$
 $21 \times 5 = 105$
 $21 \times 6 = 126$
 $21 \times 7 = 147$
 $21 \times 8 = 168$
 $21 \times 9 = 189$
 $21 \times 10 = 210$

22

$22 \times 1 = 22$
 $22 \times 2 = 44$
 $22 \times 3 = 66$
 $22 \times 4 = 88$
 $22 \times 5 = 110$
 $22 \times 6 = 132$
 $22 \times 7 = 154$
 $22 \times 8 = 176$
 $22 \times 9 = 198$
 $22 \times 10 = 220$

23

$23 \times 1 = 23$
 $23 \times 2 = 46$
 $23 \times 3 = 69$
 $23 \times 4 = 92$
 $23 \times 5 = 115$
 $23 \times 6 = 138$
 $23 \times 7 = 161$
 $23 \times 8 = 184$
 $23 \times 9 = 207$
 $23 \times 10 = 230$

24

$24 \times 1 = 24$
 $24 \times 2 = 48$
 $24 \times 3 = 72$
 $24 \times 4 = 96$
 $24 \times 5 = 120$
 $24 \times 6 = 144$
 $24 \times 7 = 168$
 $24 \times 8 = 192$
 $24 \times 9 = 216$
 $24 \times 10 = 240$

25

$25 \times 1 = 25$
 $25 \times 2 = 50$
 $25 \times 3 = 75$
 $25 \times 4 = 100$
 $25 \times 5 = 125$
 $25 \times 6 = 150$
 $25 \times 7 = 175$
 $25 \times 8 = 200$
 $25 \times 9 = 225$
 $25 \times 10 = 250$

26

$26 \times 1 = 26$
 $26 \times 2 = 52$
 $26 \times 3 = 78$
 $26 \times 4 = 104$
 $26 \times 5 = 130$
 $26 \times 6 = 156$
 $26 \times 7 = 182$
 $26 \times 8 = 208$
 $26 \times 9 = 234$
 $26 \times 10 = 260$

27

$27 \times 1 = 27$
 $27 \times 2 = 54$
 $27 \times 3 = 81$
 $27 \times 4 = 108$
 $27 \times 5 = 135$
 $27 \times 6 = 162$
 $27 \times 7 = 189$
 $27 \times 8 = 216$
 $27 \times 9 = 243$
 $27 \times 10 = 270$

28

$28 \times 1 = 28$
 $28 \times 2 = 56$
 $28 \times 3 = 84$
 $28 \times 4 = 112$
 $28 \times 5 = 140$
 $28 \times 6 = 168$
 $28 \times 7 = 196$
 $28 \times 8 = 224$
 $28 \times 9 = 252$
 $28 \times 10 = 280$

29

$29 \times 1 = 29$
 $29 \times 2 = 58$
 $29 \times 3 = 87$
 $29 \times 4 = 116$
 $29 \times 5 = 145$
 $29 \times 6 = 174$
 $29 \times 7 = 203$
 $29 \times 8 = 232$
 $29 \times 9 = 261$
 $29 \times 10 = 290$

30

$30 \times 1 = 30$
 $30 \times 2 = 60$
 $30 \times 3 = 90$
 $30 \times 4 = 120$
 $30 \times 5 = 150$
 $30 \times 6 = 180$
 $30 \times 7 = 210$
 $30 \times 8 = 240$
 $30 \times 9 = 270$
 $30 \times 10 = 300$

(6) Powers of 2: 2^2 2^{16}

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32$$

$$2^6 = 64$$

$$2^7 = 128$$

$$2^8 = 256$$

$$2^9 = 512$$

$$2^{10} = 1024$$

$$2^{11} = 2048$$

$$2^{12} = 4096$$

$$2^{13} = 8192$$

$$2^{14} = 16384$$

$$2^{15} = 32768$$

$$2^{16} = 65536$$

(7) Important Multiplication Values

$$37 \times 3 = 111$$

$$67 \times 3 = 201$$

$$27 \times 3 = 81$$

$$87 \times 3 = 261$$

(8) Powers of 3 & 5 (up to 3^{10} and 5^{10})

#1

The powers of 3 up to the 10th power are as follows:

$$3^1 = 3$$

$$3^2 = 9$$

$$3^3 = 27$$

$$3^4 = 81$$

$$3^5 = 243$$

$$3^6 = 729$$

$$3^7 = 2,187$$

$$3^8 = 6,561$$

$$3^9 = 19,683$$

$$3^{10} = 59,049$$

The powers of 5 up to 10 are as follows:

$$5^1 = 5$$

$$5^2 = 25$$

$$5^3 = 125$$

$$5^4 = 625$$

$$5^5 = 3,125$$

$$5^6 = 15,625$$

$$5^7 = 78,125$$

$$5^8 = 390,625$$

$$5^9 = 1,953,125$$

$$5^{10} = 9,765,625$$

(9) Factorials: 1! to 10!

n Factorial	Value
1 Factorial	1
2 Factorial	2
3 Factorial	6
4 Factorial	24
5 Factorial	120
6 Factorial	720
7 Factorial	5040
8 Factorial	40,320
9 Factorial	362,880
10 Factorial	3,628,800

(10) Prime numbers: Up to 100

Prime Numbers up to 100



What is a prime number?

A prime number is a number that can only be divided by itself and 1 without remainders.

1 is not a prime number because it has only one factor - itself. Prime numbers need to have exactly two factors.

Some prime number facts...



- 2 is the smallest prime number.
- 2 is the only prime number that is even.
- 2 and 3 are the only consecutive prime numbers.
- No prime number greater than 5 ends in a 5.
- Prime numbers become rarer as the numbers get larger.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100